

1.Purpose

This Risk Management Plan aims at defining a principled and complete approach to the risk management problem of identifying, evaluating, managing, and monitoring risks which would affect the achievement of the objectives of the Phishing Detection Tool with Real Time Threat Intelligence project. The purpose of this plan is to prevent as many threats as possible in advance so there is no delay in delivery of the project withing budget and quality standards. Through systematic management of risks, the project team decreases adverse effects and capitalizes upon opportunities during the project life cycle.

2.Objectives:

- **Proactive Risk Mitigation:** Prepare for and resolve issues before they get too hefty, so project disruptions stay to a minimum and things continue to flow.
- **Continuous Monitoring:** Analyze risks in regular intervals during the project lifecycle's period of time to resolve risks promptly.
- **Collaboration and Accountability:** establish a team that accepts all types of risk under the responsibility and ownership management of one team member in each domain.
- **Stakeholder Confidence:** Develop and maintain trust by implementing a good robust and transparent risk management process that includes keeping the stakeholders involved.

3. Scope

This Risk Management Plan covers the following areas:

- **Technical Components:** It includes risks of API integrations, creating GUI, data accuracy and cybersecurity threat risks.
- **Resources:** It looks at availability and sufficiency of tools, team members and needed skillset to attain project goal.
- **External Factors:** Risks from service level dependencies, regulatory changes, and external cyber are considered that could affect the project.
- **Project Schedule:** It remains about meeting deadlines and milestones to maintain the projects on track and scope.

4. Risk Management Process

4.1 Risk Identification

Risk identification is a continuous process handled by the entire project team. Key methods used include:

1. **Brainstorming:** Regular team meetings to talk through the risks, and potential solutions.
2. **Checklists:** Basing backlog with historical project data & best practices to predict the generic issues.
3. **Expert Input:** Talking to subject matter experts in cybersecurity and software development for their insights on and suggestions for cybersecurity of SaaS products.
4. **SWOT Analysis:** An internal and external SWOT analysis of factors or forces relevant to the project that may affect its success.

Risk Categories and Examples

- **Technical Risks:**
 1. Using this site is subject to API downtime or rate limit breach.
 2. Phishing detection accuracy errors due to URL/IP classification errors.
 3. GUI bugs that eliminate user experience.
- **Resource Risks:**
 1. Critical team members are unavailable during critical project phases.
 2. Inadequate experience with managing a certain set of APIs or data visualization tools.
- **External Risks:**
 1. Whether services are cut or changed due to API policy changes.
 2. Malicious URLs/IPs compromising the system impacting our website.
- **Schedule Risks:**
 1. We have foreseen that there will be slippage in the meaning of development timelines because of issues that we did not anticipate in our technical terms.
 2. Increased Fractions of late delivery which are driven by tail end phases such as the testing and validation phases.

4.2 Risk Assessment

Each risk is evaluated based on:

Likelihood (L): (1 = Low, 5 = Higher) Probability of occurrence.

Impact (I): The effect on project objectives (1 = Low, 5 = High).

Risk Score (R): $R = L \times I$.

Mitigation Difficulty (D): Ease with which mitigation strategies would be implemented (1 for easy, 5 for difficult).

Risk Assessment Table:

Risk	Priority	Impact[I]	Likelihood[L]	Risk Score[R]	Mitigation Difficulty[D]
API Downtime	Critical	5	4	20	3
Improper Input Classification	Medium	3	2	6	2
Exceeding rate limits	High	4	3	12	2
Internet Connectivity Issues	High	4	3	12	3

Cybersecurity Threats	Critical	5	3	15	4
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4.3 Risk Mitigation Strategies

Potential impacts of each of the identified risks in the project are mitigated by specific mitigation strategies. The following are the detailed mitigation strategies for each risk category along with their respective owners:

- **API Downtime:** We will integrate backup APIs and have retry mechanisms to mitigate the risk of API down time. Any unavailability of a service will be instantly notified to the users. This strategy applies in the form of mitigation by the API Integration Lead.
- **Exceeding Rate Limits:** As a measure to manage the potential of using API rate limits, we will monitor API usage in real time and make throttled requests as needed. When users reach an approaching usage limit, they will be notified. These measures are to be implemented by the Developer.
- **Incorrect Input Classification:** With regex validation we will improve the accuracy of input classification by allowing manual override of user input. Task as such come under responsibility of Team Lead.
- **Cybersecurity Threats:** In order to protect ourselves from cybersecurity threats, we will sandbox our testing environments to try, we'll sanitize all inputs and isolate any questionable behavior immediately. These activities will be conducted by the Security Analyst.
- **Internet Connectivity Issues:** In order to ensure internet connectivity, we will often synchronize project progress to GitHub and allow for partial offline functionality where possible. The Project Manager will make sure these strategies are effectively implemented.

4.4 Risk Monitoring and Control

Ongoing Monitoring:

1. Reviews of risk statuses with new risks discussed and mitigation strategies evaluated at a weekly team meeting.
2. A risk register is used to track, update, and document a risk status and its mitigation actions.

Status Indicators:

- **Open:** Some risk is identified, and the risk is to be mitigated.
- **In Progress:** Strategies are being used to mitigate.
- **Mitigated:** But it's under control and still being monitored.
- **Closed:** It has been resolved or made invalid.

Risk Register Example

Risk	Impact	Likelihood	Status	Owner	Mitigation Actions
API Downtime	5	4	In Progress	API Integration Lead	Tested backup APIs, added retry logic
Improper Input Classification	4	3	Open	Developer	Request throttling can be done.
Exceeding rate limits	5	3	Mitigated	Security Analyst	Inputs sanitized, tested established.

5. Roles and Responsibilities

Roles	Responsibility
Project Manager	Manage risk, review specific mitigation plans and converse with stakeholders.
API Integration Lead	Addressing API risks that include API downtime and usage limits, so that our API is working smoothly.
GUI Developer	Keep track of graphical interface risks and risk mitigation and fulfill the requirement of user-friendly experience by resolving every usability issue.
Security Analyst	Guard your URLs/IPs from insecure handling, cut out the risk of cybersecurity attacks, and detect malicious activities on a continuous basis.
Team Members	They look at their unique area and identify risks as well as proposed mitigation strategies and collaborate in risk management.

Detailed Responsibilities

- **Project Manager:** Coordinates risk management activities, ensures that risk mitigation strategies are well implemented; and delivers risk status to stakeholders.
- **API Integration Lead:** It monitors API performance, enacts backup solution and API rate limit is not exceeded by it.
- **GUI Developer:** It conducts usability testing and even identifies and resolves Graphical User Interface bugs and keeps a smooth user experience.

- **Security Analyst:** It tests for vulnerabilities to security protocols and educates the team on best practices for handling sensitive data.
- **Team Members:** Help the team to identify new risks, take active part in risk identification sessions, keep the risk register updated with new risks and mitigation actions, assist to deal with problems.

6. Escalation and Contingency Plans

Contingency Plans

- **API Downtime:** If there is no response use a fallback to cached data or other APIs to keep services up and running and to minimize disruption with users.
- **Rate Limits:** It raises the queue requests, introduces delays for any non-critical functionality, notifies the users about any limitations and suggests a retry interval.
- **Cybersecurity Incidents:** Log and isolate malicious activities immediately to show the slightest signs that occur before they compromise systems broadly and develop and follow through on incident response protocols to reduce the impact of activities if they occur.

Escalation Plans

- **Critical Risks:** We immediately escalate any risk rated as critical to the Project Manager and stake holders to get them addressed as forcibly as possible.
- **Regular Reporting:** Invoice key people, stakeholders on continuous communication in case the information regarding risk status and mitigation efforts are needed by all team members and stakeholders.

7. Review and Updates

- **Periodic Reviews:** The Risk Management Plan will be reviewed in the first project phase and then at regular phases of review meetings to verify its significance and usefulness. An update to the plan will happen immediately to any changes in project scope or new risks identified.
- **Documentation Updates:** The Risk Management Plan will be changed, recorded, and communicated to all team and stakeholder who need to know changes in order to be transparent and hold the whole team in alignment.
- **Continuous Improvement:** The risk management process will be informed by feedback from the team and stakeholders and will produce a culture of continuous learning and improvement.